ACADEMIC PROJECT BRIEF & TECHNICAL SUMMARY

Academic Year 2026

NearbyChat

Design and Security Analysis of a Secure Peer-to-Peer Mesh Communication System

Bluetooth Low Energy (BLE) Dual-Role | X25519 ECDH + AES-GCM-256 | Zero Infrastructure Dependency | Multi-Hop Mesh Relaying | Perfect Forward Secrecy

EXECUTIVE SUMMARY & CORE OBJECTIVE

Modern communication platforms (Signal, WhatsApp, Telegram) depend strictly on active cellular infrastructure, central servers, and global internet routing. During natural disasters, state-mandated shutdowns, or remote field operations, centralized channels collapse. **NearbyChat** resolves this vulnerability by delivering an autonomous, serverless, zero-knowledge direct and multi-hop mesh communication system. Operating over standard smartphone **Bluetooth Low Energy (BLE)** hardware, it enables end-to-end encrypted messaging with no phone numbers, email addresses, or external network infrastructure.

DUAL-DIMENSION IMPLEMENTATION

Dimension 1: Mobile Client Application

Flutter 3.x • Dart • Android 7.0+ & iOS 13.0+

Full-featured mobile client with simultaneous BLE Central & Peripheral GATT operations. Enables seamless peer discovery, instant message transmission, Safety Number verification, and on-device hardware-backed encrypted storage.

Dimension 2: Research & Mesh Simulation

Python 3.10+ • Mininet-WiFi • Linux Namespaces

Rigorous academic emulation platform modeling 4–10 virtual wireless nodes. Evaluates multi-hop packet routing, dynamic route failover, dropper node isolation, trust scoring heuristics, and replay attack immunity under network stress.

CRYPTOGRAPHIC MODEL

- **X25519 ECDH Key Agreement:** Hardware-backed Diffie-Hellman session secret derivation per conversation.
- **AES-GCM-256 Authenticated Cipher:** Authenticated ciphertexts with random 12-byte IVs guaranteeing confidentiality and tampering detection.
- **Safety Numbers:** SHA-256 fingerprint verification (6 numeric blocks) preventing Man-in-the-Middle (MITM) attacks.
- **Authenticated HMAC ACKs:** Cryptographic delivery receipts verified via session keys to eliminate spoofing.
- **At-Rest Encryption:** Local chat histories fully encrypted before persistent storage on disk.

NETWORK & PROTOCOL DESIGN

- **Simultaneous Dual-Role GATT:** Simultaneously broadcasts BLE advertising packets and scans for nearby peers.
- **MTU Adaptive Slicing:** Binary frame codec splitting arbitrarily large payloads into 20–512 byte fragments with CRC32 integrity.
- **Multi-Hop Mesh Routing:** Dynamic next-hop routing tables, hop limits (TTL=7), and duplicate message suppression.
- **Anti-Spoofing & Replay Defense:** Nonce tracking and public key-to-username binding prevent replay and impersonation.
- **Automated Test Verification:** 28/28 Flutter unit tests + 9/9 Python pytest tests passing (100% test pass rate).

EMPIRICAL VERIFICATION & PERFORMANCE METRICS

100%	< 1.2s	256-bit	0 B
AUTOMATED TESTS PASSED (37/37)	PEER DISCOVERY LATENCY	AES-GCM AUTHENTICATED CRYPTO	PLAINTEXT METADATA LEAKED

PROJECT AUTHORS & TEAM MEMBERS

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Source Code & Complete Repository
Full implementation, academic research papers, test suites, and interactive web demos

OFFICIAL GITHUB REPOSITORY
https://github.com/Yash-Maholiya/NearbyChat